

EE6333: Homework 3

Bayesian Estimation

Prof. Sundeep Rangan

1. (a) The MAP estimate is

$$\hat{x} = \arg \max_x p(x|y) = \arg \max_x C(y)e^{-H(x,y)} = \arg \min_x H(x,y),$$

where we have used the fact that $C(y)$ does not depend on x .

- (b) Using the fact that the density must integrate to one,

$$1 = \int p(x|y)dx = C(y) \int e^{-H(x,y)}dx = C(y)Z_0.$$

Hence, $C(y) = 1/Z_0$. Therefore,

$$\mathbb{E}(x|y) = \int xp(x|y)dx = \frac{1}{Z_0} \int xe^{-H(x,y)}dx = \frac{Z_1}{Z_0}.$$

Similarly,

$$\mathbb{E}(x^2|y) = \int x^2p(x|y)dx = \frac{1}{Z_0} \int x^2e^{-H(x,y)}dx = \frac{Z_2}{Z_0},$$

and hence

$$\text{var}(x|y) = \mathbb{E}(x^2|y) - \mathbb{E}^2(x|y) = \frac{Z_2}{Z_0} - \frac{Z_1^2}{Z_0^2}.$$

- (c) Given the joint density $p(x, y)$ the posterior is

$$p(x|y) = \frac{p(x, y)}{p(y)} = \frac{C(y)}{p(y)}e^{-H(x,y)} = C'(y)e^{-H(x,y)},$$

where $C'(y) = C(y)/p(y)$. Hence, posterior $p(x|y)$ is in the same form as in part (a), only with a different function $C(y)$.

2. (a) Since this is an AWGN channel, the ML estimate is

$$\hat{\theta} = \arg \min_{\theta \geq 0} (x - \theta)^2.$$

Since the minimization is to be performed over $\theta \geq 0$, we have the solution

$$\hat{\theta} = \begin{cases} 0 & \text{if } x \leq 0 \\ x & \text{if } x \geq 0. \end{cases}$$

Equivalently, we can write $\theta = \max\{x, 0\}$.

(b) From Bayes' rule, the posterior density is

$$p(\theta|x) = \frac{1}{p(x)}p(x|\theta)p(\theta) = \frac{1}{p(x)\sqrt{2\pi}\sigma\lambda} \exp \left[-\frac{(x-\theta)^2}{2\sigma^2} - \frac{\theta}{\lambda} \right]. \quad (1)$$

Hence, we can take

$$H(\theta, x) = \frac{(x-\theta)^2}{2\sigma^2} + \frac{\theta}{\lambda}, \quad C(x) = \frac{1}{p(x)\sqrt{2\pi}\sigma\lambda}.$$

(c) From the previous problem

$$\hat{\theta} = \arg \min_{\theta \geq 0} H(\theta, x).$$

When we have hard constraint of the form $\theta \geq 0$, we break this minimization into two cases: Either $\theta > 0$ or $\theta = 0$. If $\theta > 0$, then the minimum is in an open neighborhood of the feasible set, so $\partial H(\theta, x)/\partial \theta = 0$ which implies

$$\frac{\theta - x}{\sigma^2} + \frac{1}{\lambda} = 0 \Rightarrow \hat{\theta} = x - \frac{\sigma^2}{\lambda}.$$

In order that $\hat{\theta} > 0$, we need that $x \geq \sigma^2/\lambda$. If this condition is not satisfied, then there cannot be a minima with $\hat{\theta} > 0$. Hence, we must have $\hat{\theta} = 0$. We conclude that the MAP estimate is

$$\hat{\theta} = \begin{cases} 0 & \text{if } x \leq t \\ x - t & \text{if } x \geq t, \end{cases} \quad t = \sigma^2/\lambda.$$

(d) Define

$$\mu = x - \frac{\sigma^2}{\lambda},$$

so that

$$\frac{(\theta - x)^2}{2\sigma^2} + \frac{\theta}{\lambda} = \frac{(\theta - \mu)^2}{2\sigma^2} + \frac{x^2 - t^2}{2\sigma^2}.$$

Hence, from (1), for $\theta > 0$,

$$p(\theta|x) = \frac{1}{p(x)\sqrt{2\pi}\sigma\lambda} \exp \left[\frac{x^2 - t^2}{2\sigma^2} \right] \exp \left[-\frac{(\theta - \mu)^2}{2\sigma^2} \right] = C(x)\phi \left(\frac{(\theta - \mu)^2}{\sigma^2} \right),$$

where $C(x)$ is a function that depends only on x (and not θ) and $\phi(u)$ is the pdf of the standard Gaussian

$$\phi(u) = \frac{1}{\sqrt{2\pi}} e^{-u^2/2}. \quad (2)$$

Since (2) is restricted to $\theta > 0$, this is identical the truncated normal distribution $\theta \sim \mathcal{N}(\mu, \sigma)$ restricted to $\theta \in [a, b]$ where $a = 0$ and $b = \infty$. From the wikipedia page the posterior mean is

$$\mathbb{E}(\theta|x) = \mu + \frac{\phi(u)\sigma}{Q(u)},$$

where $u = -\mu/\sigma$.

(e) We can compute the estimates with the following code:

```
% Parameters
lam = 1;           % Exp(theta)
wvar = 0.1;        % noise variance
nx = 200;          % number of points
x = linspace(-2,3,nx)'; % points to test

% ML estimate
xml = max(0,x);

% MAP estimate
t = wvar/lam;
mu = x-t;
xmap = max(0, mu);

% Compute MMSE estimate
u = -mu/sqrt(wvar);
phiu = 1/sqrt(2*pi)*exp(-u.^2/2);
xmmse = mu + sqrt(wvar)*phiu./qfunc(u);

% Plot the results
plot(x,[xml xmap xmmse], '-','Linewidth',2);
grid on;
set(gca,'FontSize',16);
legend('ML','MAP','MMSE','Location','NorthWest');
xlabel('x');
ylabel('thetaHat');
```

The results are shown in Fig. 1.

3. The expected loss is

$$J(\hat{\theta}) = E(L(\theta, \hat{\theta})|x) = \int_{-\infty}^{\infty} |\theta - \hat{\theta}| p(\theta|x) d\theta.$$

We break this integral into two parts,

$$J(\hat{\theta}) = \int_{\hat{\theta}}^{\infty} (\theta - \hat{\theta}) p(\theta|x) d\theta + \int_{-\infty}^{\hat{\theta}} (\hat{\theta} - \theta) p(\theta|x) d\theta.$$

Using Leibnitz rule

$$\begin{aligned} \frac{\partial J(\hat{\theta})}{\partial \hat{\theta}} &= -(\hat{\theta} - \hat{\theta}) p(\hat{\theta}|x) - \int_{\hat{\theta}}^{\infty} p(\theta|x) d\theta + (\hat{\theta} - \hat{\theta}) p(\hat{\theta}|x) + \int_{-\infty}^{\hat{\theta}} p(\theta|x) d\theta \\ &= -P(\theta > \hat{\theta}) + P(\theta \leq \hat{\theta}). \end{aligned}$$

Setting the derivative to zero, we obtain that

$$P(\theta > \hat{\theta}) = P(\theta \leq \hat{\theta}),$$

which implies that $\hat{\theta}$ is the median of the density, $p(\theta|x)$.

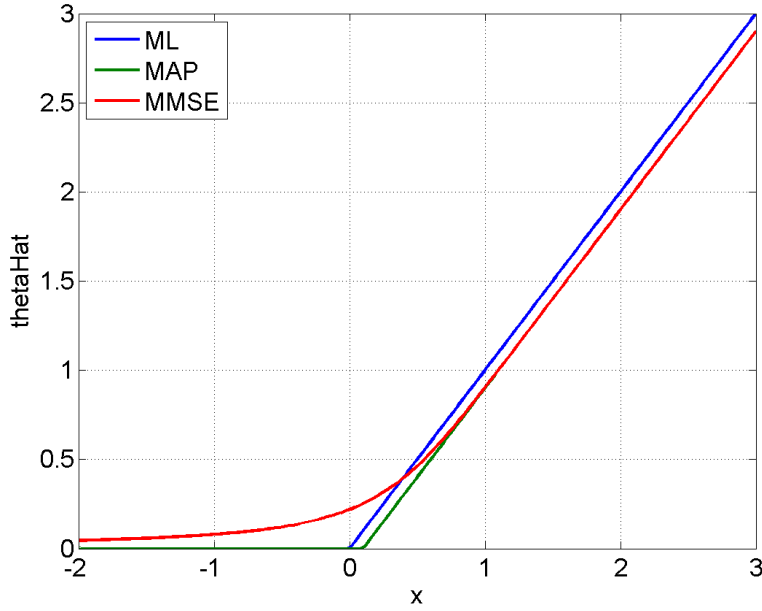


Figure 1: Estimators for Problem 2.

4. (a) The negative log likelihood is

$$p(\mathbf{x}|\theta) = \prod_{i=1}^N p(x_i|\theta) = \theta^N \prod_{i=1}^N e^{-\theta x_i},$$

so the negative log likelihood is

$$L(\theta) = -\ln p(\mathbf{x}|\theta) = -N \ln(\theta) + \theta \sum_{i=1}^N x_i.$$

Setting $L'(\theta) = 0$, we obtain

$$-\frac{N}{\theta} + \sum_{i=1}^N x_i \Rightarrow \hat{\theta} = \left[\frac{1}{N} \sum_{i=1}^N x_i \right]^{-1}.$$

- (b) The Gamma distribution is

$$p(\theta) = \frac{\beta^\alpha}{\Gamma(\alpha)} \theta^{\alpha-1} e^{-\beta\theta} \propto \theta^{\alpha-1} e^{-\beta\theta},$$

where the proportional to sign ($\propto$) indicates a constant that does not depend on θ . Hence,

$$p(\theta|\mathbf{x}) \propto p(\mathbf{x}|\theta)p(\theta) \propto \theta^N \prod_{i=1}^N e^{-x_i\theta} \theta^{\alpha-1} e^{-\beta\theta} = \theta^{\alpha'-1} e^{-\beta'\theta},$$

where

$$\alpha' = \alpha + N, \quad \beta' = \beta + \sum_{i=1}^N x_i.$$

- (c) Let $\mu = \mathbb{E}(\theta)$, and $\tau = \text{var}(\theta)$ be the prior mean and variance. We know from the wikipedia page on the Gamma distribution,

$$\mu = \frac{\alpha}{\beta}, \quad \tau = \frac{\alpha}{\beta^2} \Rightarrow \beta = \frac{\mu}{\tau}, \alpha = \frac{\mu^2}{\tau}.$$

Hence, the posterior mean is

$$\mathbb{E}(\theta|\mathbf{x}) = \frac{\alpha'}{\beta'} = \frac{\alpha + N}{\beta + N\bar{x}} = \frac{\mu^2 + N\tau}{\mu + N\bar{x}\tau},$$

where $\bar{x} = (1/N) \sum_i x_i$ is the sample mean. The posterior variance is

$$\text{var}(\theta|\mathbf{x}) = \frac{\alpha'}{(\beta')^2} = \frac{\alpha + N}{(\beta + N\bar{x})^2} = \tau \frac{\mu^2 + N\tau}{(\mu + N\bar{x}\tau)^2}.$$

- (d) As $\tau \rightarrow \infty$,

$$\mathbb{E}(\theta|\mathbf{x}) \rightarrow \frac{1}{\bar{x}},$$

which is the ML estimate. This makes sense: as $\tau \rightarrow \infty$, the prior becomes non-informative so the MMSE estimate converges to the MLE.